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MIL-M-28787/528
17 May 1991

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MIL-STD-1397 TYPE D/E SERIAL CONTROL MODULE, KEY CODE AJK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic MIL-STD-1397 type D/E serial control module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 2.573 failures per million operating hours for class I or 7.778 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be AJK.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ∓ 5°C		Class II 4/ -55 ± 5°C 85 ∓ 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
A2-A41	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
B1-B6	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vpp stress	IV	S	----------Module shall pass all remaining tests----------						
D1	Ibd	V	S	-0.12	0.22	-0.12	0.22	-0.12	0.22	mA
D2-D5	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D6	Ibd	V	S	-0.12	0.22	-0.12	0.22	-0.12	0.22	mA
D7	Ibd	V	S	-0.12	1.11	-0.12	1.11	-0.12	1.11	mA
D8,D9	Ibd	V	S	-0.12	0.22	-0.12	0.22	-0.12	0.22	mA
D10-D15	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D16	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D17	Ibd	V	S	0.0	4.0	0.0	4.0	0.0	4.0	mA
D18-D24	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
E1	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E2	Ios	VI	S	-220.0	-80.0	-220.0	-80.0	-220.0	-80.0	mA
E3	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E4-E8	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E9	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E10-E25	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
F1	Ith	VII	X	-5.73	-2.85	-5.73	-2.85	-5.73	-2.85	mA
F2	Ith	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F3	Ith	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F4	Ith	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F5	Ith	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F6	Ith	VII	X	-5.63	-2.85	-5.63	-2.85	-5.63	-2.85	mA
F7	Ith	VII	X	-7.43	-2.92	-7.43	-2.82	-7.43	-2.82	mA
F8	Ith	VII	X	-5.63	-2.85	-5.63	-2.85	-5.63	-2.85	mA
F9	Ith	VII	X	-5.83	-2.85	-5.83	-2.85	-5.83	-2.85	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F10-F12	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F13	Iih	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F14,F15	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F16	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F17	Iih	VII	X	-8.0	0.2	-8.0	0.2	-8.0	0.2	mA
F18-F24	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F25-F40	Iih	VII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
G1	Iil	VIII	X	-5.73	-2.85	-5.73	-2.85	-5.73	-2.85	mA
G2	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G3	Iil	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G4	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G5	Iil	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G6	Iil	VIII	X	-5.83	-2.85	-5.83	-2.85	-5.83	-2.85	mA
G7	Iil	VIII	X	-7.43	-2.82	-7.43	-2.82	-7.43	-2.82	mA
G8	Iil	VIII	X	-5.63	-2.85	-5.63	-2.85	-5.63	-2.85	mA
G9	Iil	VIII	X	-5.83	-2.85	-5.83	-2.85	-5.83	-2.85	mA
G10-G12	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G13	Iil	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G14,G15	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G16	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G17	Iil	VIII	X	-8.0	0.2	-8.0	0.2	-8.0	0.2	mA
G18-G24	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G25-G40	Iil	VIII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
H1-H25	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J26	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K	Iozh	XI	X	0.0	0.25	0.0	0.25	0.0	0.25	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
L1-L4	Vio	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L5-L7	Vio	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L8	Vio	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L9-L19	Vio	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
M1	PSFC	XIII	S	4.3	8.0	4.3	8.0	4.3	8.0	uF
M2	PSFC	XIII	S	3.7	6.9	3.7	6.9	3.7	6.9	uF
N	Icc	XIV	X	342	1487	342	1487	342	1487	mA
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
Q3,Q4	Dynamic	XVI	Q	See table XVI						
R1	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R2-R5	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R6	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R7,R8	C1/D	XVIII	Q	4	100	4	100	4	100	pF
R9	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R10-R15	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R16	C1/D	XVIII	Q	2	100	2	100	2	100	pF
R17	C1/D	XVIII	Q	8	150	8	150	8	150	pF
R18-R24	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R25-R32	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R33	C1/D	XVIII	Q	4	100	4	100	4	100	pF
R34	C1/D	XVIII	Q	4	125	4	125	4	125	pF
R35-R40	C1/D	XVIII	Q	4	100	4	100	4	100	pF
T1,T2	PSTR	XIX	Q	0	400	0	400	0	400	mV

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/528-1 for class I and M28787/528-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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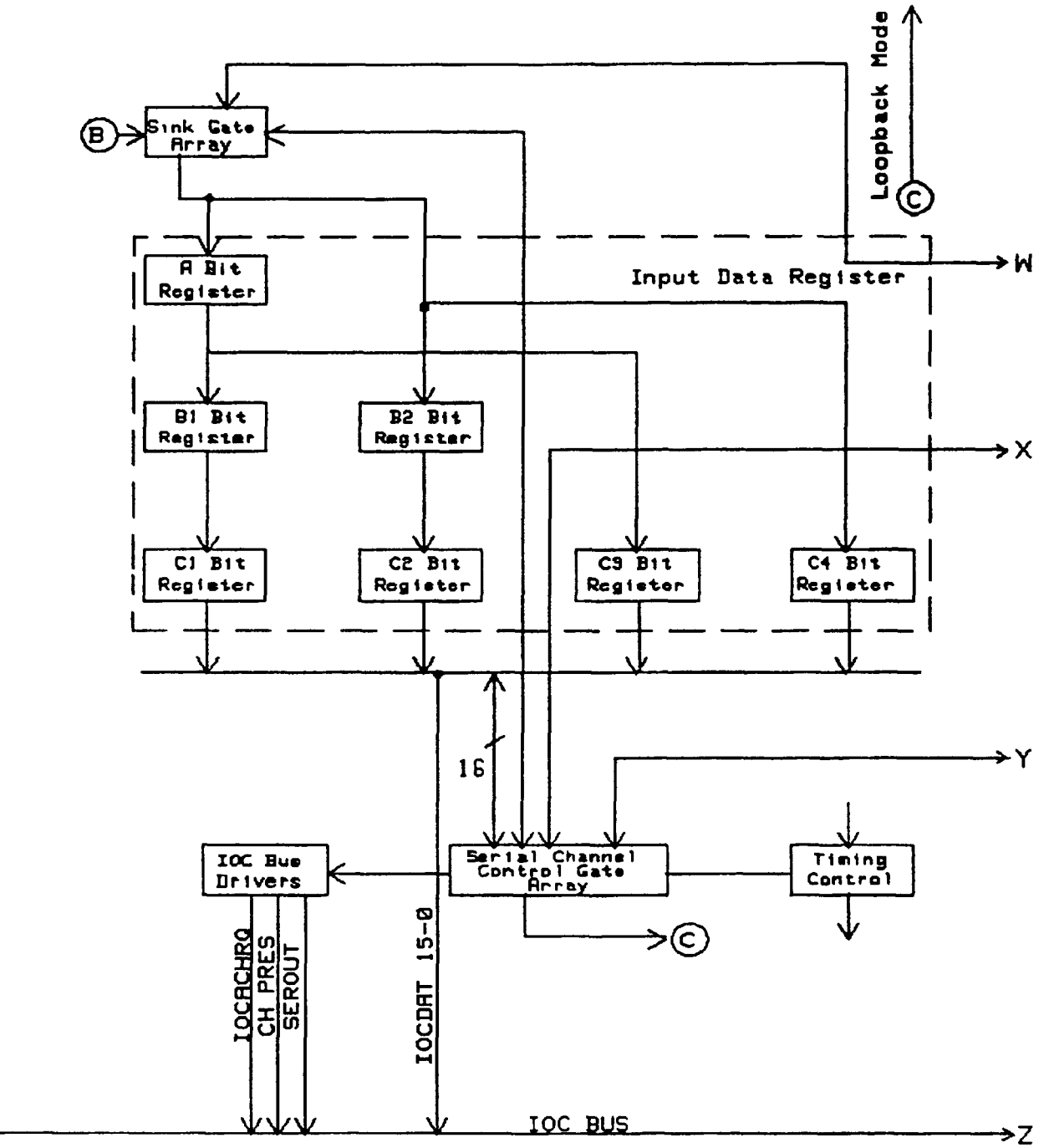


FIGURE 1. AJK functional block diagram.

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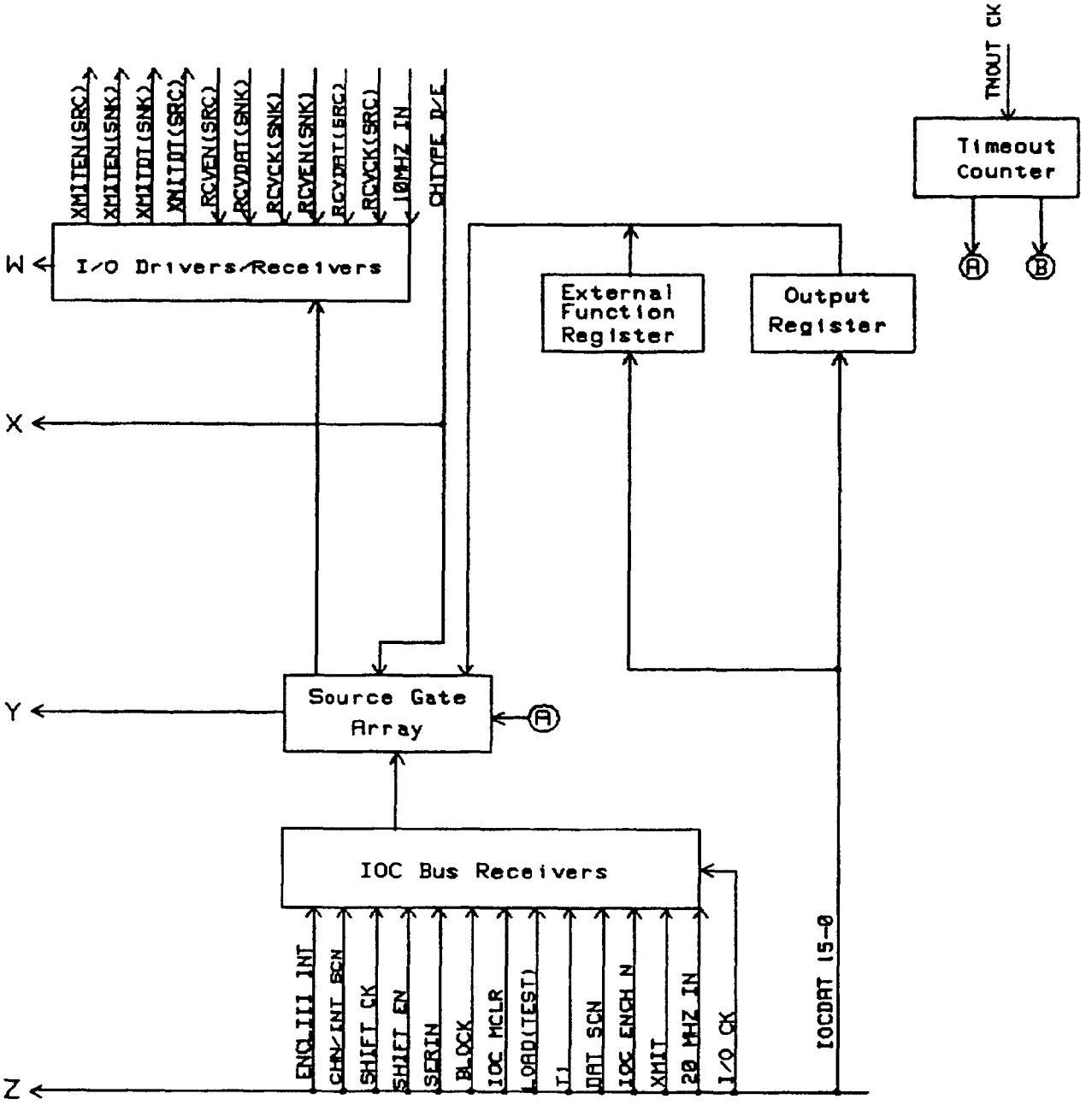


FIGURE 1. AJK functional block diagram - Continued.

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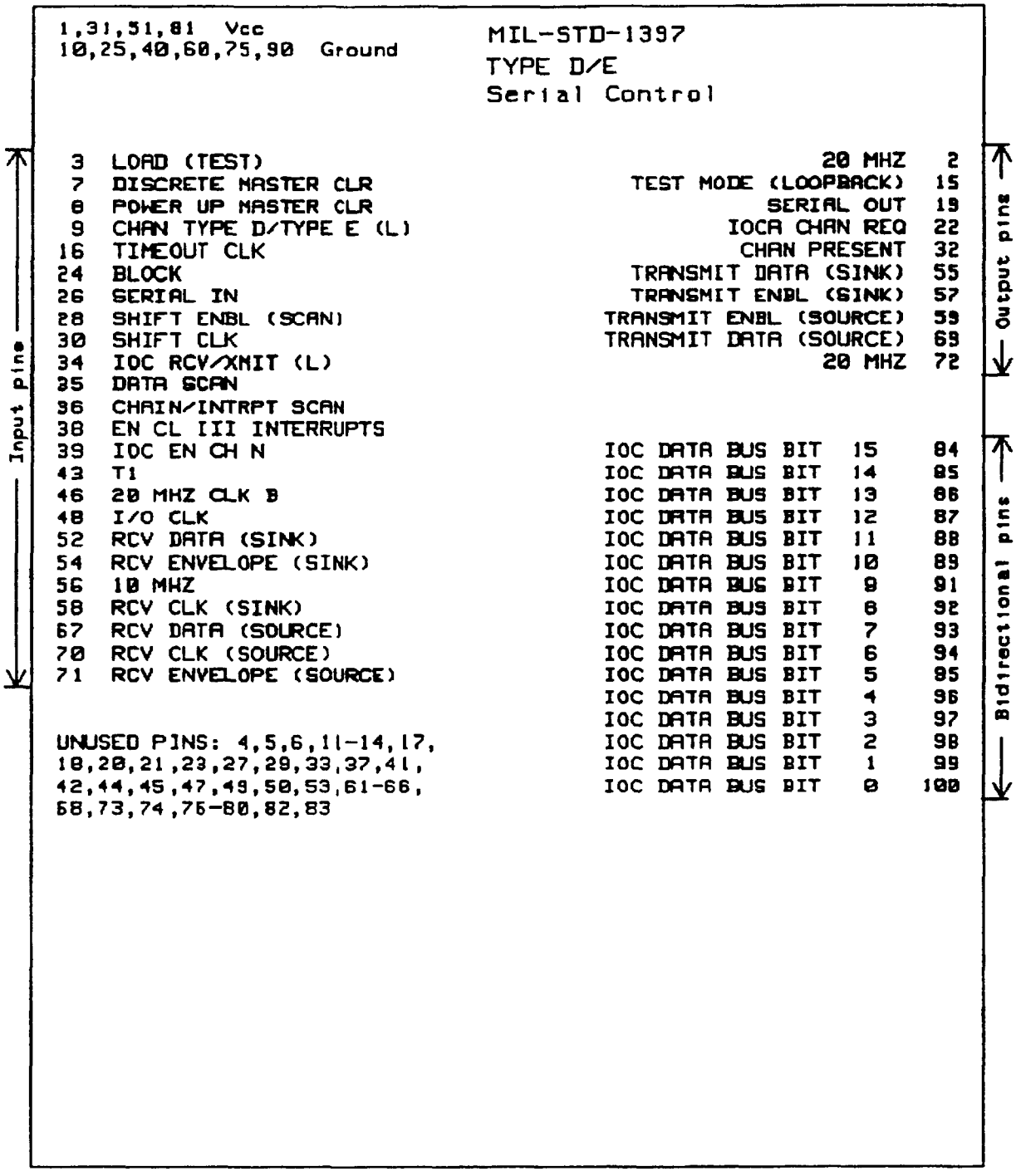


FIGURE 2. AJK pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

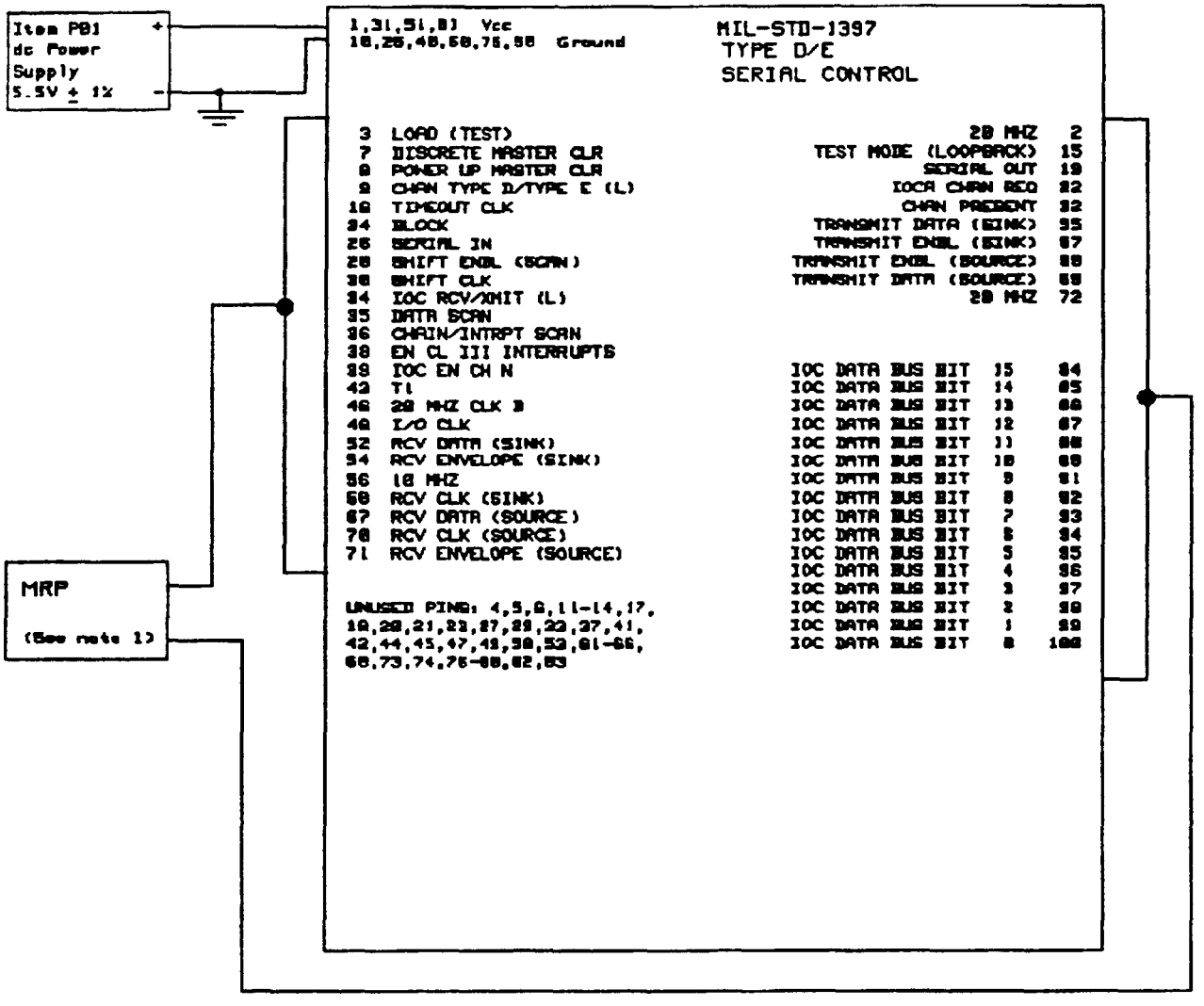
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. AJK operating temperature and component temperature test circuit.

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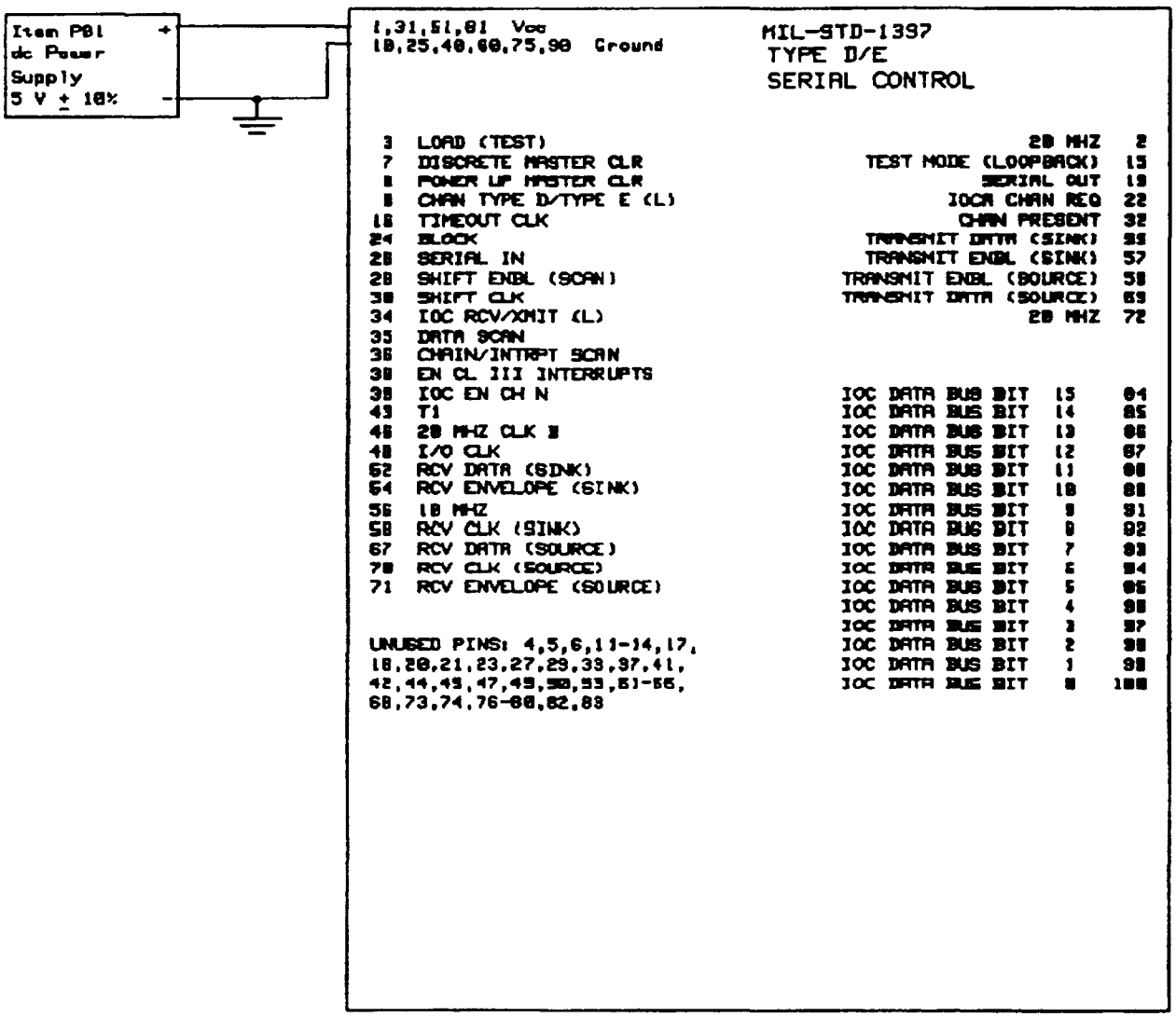


FIGURE 4. AJK vibration test circuit.

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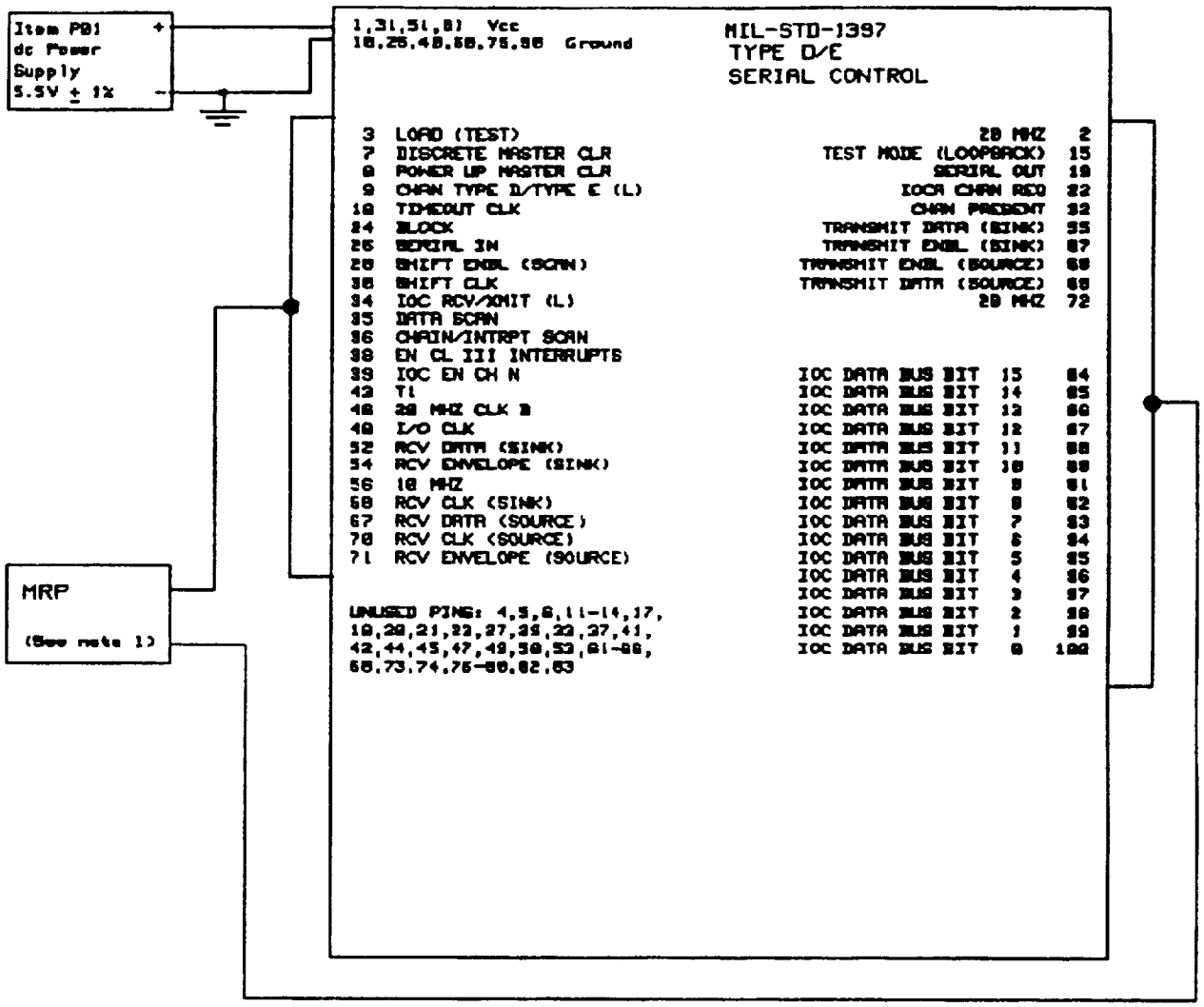
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. AJK life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 8.2 W (max); 4.9 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
LOAD(TEST)	NA	0.4	2.4	-5.73	-2.85	0.22	75
IOC MCLR	-1.5	0.4	2.4	-0.2	0.02	0.1	75
PWR MCLR	-1.5	0.4	2.4	-0.4	0.02	0.1	75
CHTYPE D/E	-1.5	0.4	2.4	-0.2	0.02	0.1	75
TMOUT CK	-1.5	0.4	2.4	-0.4	0.02	0.1	75
BLOCK	NA	0.4	2.4	-5.63	-2.85	0.22	75
SERIN	NA	0.4	2.4	-7.43	-2.82	1.11	100
SHIFT EN	NA	0.4	2.4	-5.63	-2.85	0.22	100
SHIFT CK	NA	0.4	2.4	-5.83	-2.85	0.22	75
XMIT	-1.2	0.4	2.4	-0.6	0.02	0.1	75
DAT SCN	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CHN/INT SCN	-1.2	0.4	2.4	-0.6	0.02	0.1	75
ENCLIII INT	-1.5	0.4	2.4	-0.4	0.02	0.1	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
IOC ENCH N	-1.2	0.4	2.4	-0.6	0.02	0.1	75
T1	-1.2	0.4	2.4	-0.6	0.02	0.1	75
20MHZ IN	-1.2	0.4	2.4	-2.0	0.05	1.0	100
I/O CK	-1.2	0.4	2.4	-8.0	0.2	4.0	150
RCVDAT(SNK)	-1.2	0.4	2.4	-2.0	0.05	1.0	75
RCVEN(SNK)	-1.2	0.4	2.4	-2.0	0.05	1.0	75
10MHZ IN	-1.2	0.4	2.4	-2.0	0.05	1.0	75
RCVCK(SNK)	-1.2	0.4	2.4	-2.0	0.05	1.0	75
RCVDAT(SRC)	-1.2	0.4	2.4	-2.0	0.05	1.0	75
RCVCK(SRC)	-1.2	0.4	2.4	-2.0	0.05	1.0	75
RCVEN(SRC)	-1.2	0.4	2.4	-2.0	0.05	1.0	75
IOCDAT 15	NA	0.4	2.4	-0.25	0.07	NA	75
IOCDAT 14	NA	0.4	2.4	-0.25	0.07	NA	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
IOCDAT 13	NA	0.4	2.4	-0.25	0.07	NA	75
IOCDAT 12	NA	0.4	2.4	-0.25	0.07	NA	75
IOCDAT 11	NA	0.4	2.4	-0.25	0.07	NA	75
IOCDAT 10	NA	0.4	2.4	-0.25	0.07	NA	75
IOCDAT 9	NA	0.4	2.4	-0.25	0.07	NA	75
IOCDAT 8	NA	0.4	2.4	-0.25	0.07	NA	75
IOCDAT 7	NA	0.4	2.4	-0.25	0.07	NA	100
IOCDAT 6	NA	0.4	2.4	-0.25	0.07	NA	125
IOCDAT 5	NA	0.4	2.4	-0.25	0.07	NA	100
IOCDAT 4	NA	0.4	2.4	-0.25	0.07	NA	100
IOCDAT 3	NA	0.4	2.4	-0.25	0.07	NA	100
IOCDAT 2	NA	0.4	2.4	-0.25	0.07	NA	100
IOCDAT 1	NA	0.4	2.4	-0.25	0.07	NA	100
IOCDAT 0	NA	0.4	2.4	-0.25	0.07	NA	100

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:36Z
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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
20MHZ OUT 1	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
TSTMDE(LPB)	2.4	0.5	-0.16	1.6	NA	NA	-60.0	NA
SEROUT	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
IOCACHRQ	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
XMITDT(SNK)	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
XMITEN(SNK)	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
XMITEN(SRC)	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
XMITDT(SRC)	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
20MHZ OUT 2	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
IOCDAT 15	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOCDAT 14	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOCDAT 13	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOCDAT 12	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75

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Check the source to verify that this is the current version before use.

6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
IOCDAT 11	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOCDAT 10	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOCDAT 9	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOCDAT 8	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOCDAT 7	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
IOCDAT 6	2.4	0.5	-0.4	6.0	NA	NA	-60.0	125
IOCDAT 5	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
IOCDAT 4	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
IOCDAT 3	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
IOCDAT 2	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
IOCDAT 1	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
IOCDAT 0	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
CH PRES	NA	0.5	NA	1.6	0.25	NA	NA	NA

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	10MHZ IN to XMITEN(SRC)	-	-	-	90	90	90	nS	11
1	tpd	10MHZ IN to XMITDT(SRC)	-	-	-	90	90	90	nS	11
1	tpd	10MHZ IN to XMITEN(SNK)	-	-	-	90	90	90	nS	11
1	tpd	10MHZ IN to XMITDT(SNK)	-	-	-	90	90	90	nS	11
2	tpd	10MHZ IN to SEROUT	-	-	-	85	85	85	nS	11
3	tpd	CHN/INT SCN to IOCACHRQ	-	-	-	69	69	69	nS	11
3	tpd	DAT SCN to IOCACHRQ	-	-	-	69	69	69	nS	11

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6.4 Module function. The MIL-STD-1397 type D/E serial channel control module contains all the logic to control two MIL-STD-1397 type D or E serial interface channels. The module contains a serial channel control gate array, a send and source gate array, Input/Output Controller (IOC) bus drivers and receivers, Input/Output (I/O) drivers and receivers, input register, output register and External Function (EF) register.

6.4.1 Serial channel control gate array. The serial channel control gate array is designed to provide the control functions, registers, and timing signals necessary to interface between the IOC and the sink and source functions of the MIL-STD-1397 type D/E serial interface. The gate array performs I/O buffer activities, chaining, command transfer functions, data transfer functions, and error handling. During command transfers, the gate array receives channel command codes which are used to set up data transfer parameters and to read status words. During data transfers, the gate array receives data requests from the source and sink functions. The gate array then makes a data request during a data scan. When the channel is enabled by the IOC, it sends control information to the IOC defining the type of data, and sends data if it is an input request. The IOC then sends an acknowledge code to the device and sends data if it is an output request. The gate array makes a chain/interrupt request during a chain/interrupt scan if it has a chain request or an interrupt request active. If the channel is enabled, it sends control information to the IOC which defines the type of request.

6.4.2 Drivers and receivers. The IOC bus drivers and receivers are constantly enabled and take information from and send information to the IOC bus. The I/O drivers and receivers transfer data to and from the MIL-STD-1397 type D or E serial interface cards.

6.4.3 Registers. The input register is a 32-bit 3-rank register which holds data from the sink gate array destined for the IOC bus. The output register is a 32-bit register which holds data from the IOC bus destined for the source gate array. The external function register is a 32-bit register which holds EF data destined for the source data array.

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6.4.4 Sink gate array. The sink gate array is designed to receive data words, reformat them to parallel, set requests (Input Data Request (IDR) and External Interrupt Request (EIR)), detect errors and provide channel status. For sink operation, the gate array receives serial data from the I/O receiver connected to the interface. The gate array assembles the serial data into 8-bit bytes and places them on an 8-bit data bus along with control signals used to load them in the proper registers. When a frame is received, the gate array strips off the first two bits for internal use and assembles the next 8 bits in an internal shift register. When this first byte is complete, it is placed on the data bus and a control signal is sent to load the data in the A bit register. The gate array continues to assemble the incoming data and when the second byte is complete, it is placed on the bus with a control signal which loads it into the B2 bit register. Simultaneously, the data from the A bit register is loaded into the B1 bit register. When the third byte is complete and placed on the bus, a control signal is sent that causes the following four data transfers to occur:

(B1 bit register) --> C1 bit register
(B2 bit register) --> C2 bit register
(A bit register) --> C3 bit register
(Bus) --> C4

At this point, the entire 32-bit data word is stored in the (32-bit) input register. A request is presented to the MIL-STD-1397 D/E serial channel control gate array to remove the word from this register. In overlap, another word may be received before the current one is removed from the register. In this case the first three types of the second word will be handled in the same fashion as the first word, but when the last byte is complete, the signal to load all 4 bytes into the C register will not be generated. Under these conditions, the gate array will not allow any new data words to be received and will idle until the first word is removed from the C register. When the C register is free, the gate array sends the control signal to load the C register with the second word and another request is sent to the surrounding logic.

6.4.5 Source gate array. The source gate array is designed to transmit data words, set requests (Output Data Request (ODR) and External Function Request (EFR)), and detect errors. For source operation, the gate array receives 32-bit output data or 32-bit EF data from their respective registers and sends it to the I/O driver connected to the interface.

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6.4.6 Description of I/O pins.

Signal name	Pin number	I/O	Description
LOAD(TEST)	3	I	LOAD (TEST)
IOC MCLR	7	I	DISCRETE MASTER CLR
PWR MCLR	8	I	POWER UP MASTER CLR
CHTYPE D/E	9	I	CHAN TYPE D/TYPE E (L)
TMOUT CK	16	I	TIMEOUT CLK
BLOCK	24	I	BLOCK
SERIN	26	I	SERIAL IN
SHIFT EN	28	I	SHIFT ENBL (SCAN)
SHIFT CK	30	I	SHIFT CLK
XMIT	34	I	IOC RCV/XMIT (L)
DAT SCN	35	I	DATA SCAN
CHN/INT SCN	36	I	CHAIN/INTRPT SCAN
ENCLIII INT	38	I	EN CL III INTERRUPTS
IOC ENCH N	39	I	IOC EN CH N
T1	43	I	T1
20MHZ IN	46	I	20 MHZ CLK B
I/O CK	48	I	I/O CLK
RCVDAT(SNK)	52	I	RCV DATA (SINK)
RCVEN(SNK)	54	I	RCV ENVELOPE (SINK)
10MHZ IN	56	I	10 MHZ

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6.4.6 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
RCVCK(SNK)	58	I	RCV CLK (SINK)
RCVDAT(SRC)	67	I	RCV DATA (SOURCE)
RCVCK(SRC)	70	I	RCV CLK (SOURCE)
RCVEN(SRC)	71	I	RCV ENVELOPE (SOURCE)
20MHZ OUT 1	2	O	20 MHZ
TSTMDE(LPB)	15	O	TEST MODE (LOOPBACK)
SEROUT	19	O	SERIAL OUT
IOCACHRQ	22	O	IOCA CHAN REQ
XMITDT(SNK)	55	O	TRANSMIT DATA (SINK)
XMITEN(SNK)	57	O	TRANSMIT ENBL (SINK)
XMITEN(SRC)	59	O	TRANSMIT ENBL (SOURCE)
XMITDT(SRC)	69	O	TRANSMIT DATA (SOURCE)
20MHZ OUT 2	72	O	20 MHZ
IOCDAT 15	84	I/O	IOC DATA BUS BIT 15
IOCDAT 14	85	I/O	IOC DATA BUS BIT 14
IOCDAT 13	86	I/O	IOC DATA BUS BIT 13
IOCDAT 12	87	I/O	IOC DATA BUS BIT 12
IOCDAT 11	88	I/O	IOC DATA BUS BIT 11
IOCDAT 10	89	I/O	IOC DATA BUS BIT 10
IOCDAT 9	91	I/O	IOC DATA BUS BIT 9

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6.4.6 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
IOCDAT 8	92	I/O	IOC DATA BUS BIT 8
IOCDAT 7	93	I/O	IOC DATA BUS BIT 7
IOCDAT 6	94	I/O	IOC DATA BUS BIT 6
IOCDAT 5	95	I/O	IOC DATA BUS BIT 5
IOCDAT 4	96	I/O	IOC DATA BUS BIT 4
IOCDAT 3	97	I/O	IOC DATA BUS BIT 3
IOCDAT 2	98	I/O	IOC DATA BUS BIT 2
IOCDAT 1	99	I/O	IOC DATA BUS BIT 1
IOCDAT 0	100	I/O	IOC DATA BUS BIT 0
CH PRES	32	O	CHAN PRESENT

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0048)

Agent: NW

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	C,4,5, 6,11, 12,13, 14,17, 18,20, 21,23, 27,29, 33,37, 41,42, 44,45, 47,49, 50,53, 61,62, 63,64, 65,66, 68,73, 74,76, 77,78, 79,80, 82,83	Frame	10	INF	MOhm	1-9
A2	Isolation, used to unused pins	C	4,5,6, 11,12, 13,14, 17,18, 20,21, 23,27, 29,33, 37,41, 42,44, 45,47, 49,50, 53,61, 62,63, 64,65, 66,68, 73,74, 76,77, 78,79, 80,82, 83	10	INF	MOhm	1-9

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A3-A41	Isolation, unused to unused pins	See note 4		10	INF	MOhm	1-9

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1-A41
Isolation measurement	NA	± 500 KOhm	A1-A41

2. The ohmmeter used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. The unused pins on this module are: 4,5,6,11,12,13,14,17, 18,20,21,23,27,29,33,37,41,42,44,45,47,49,50,53,61,62,63,64,65, 66,68,73,74,76,77,78,79,80,82,83. When testing for unused to unused pin isolation, connect one unused pin to A and the remaining unused pins to B.

5. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

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NOTES: - Continued.

6. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

7. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

8. The test circuit shall be as indicated on figure 6.

9. Abbreviations used are defined in 6.1.

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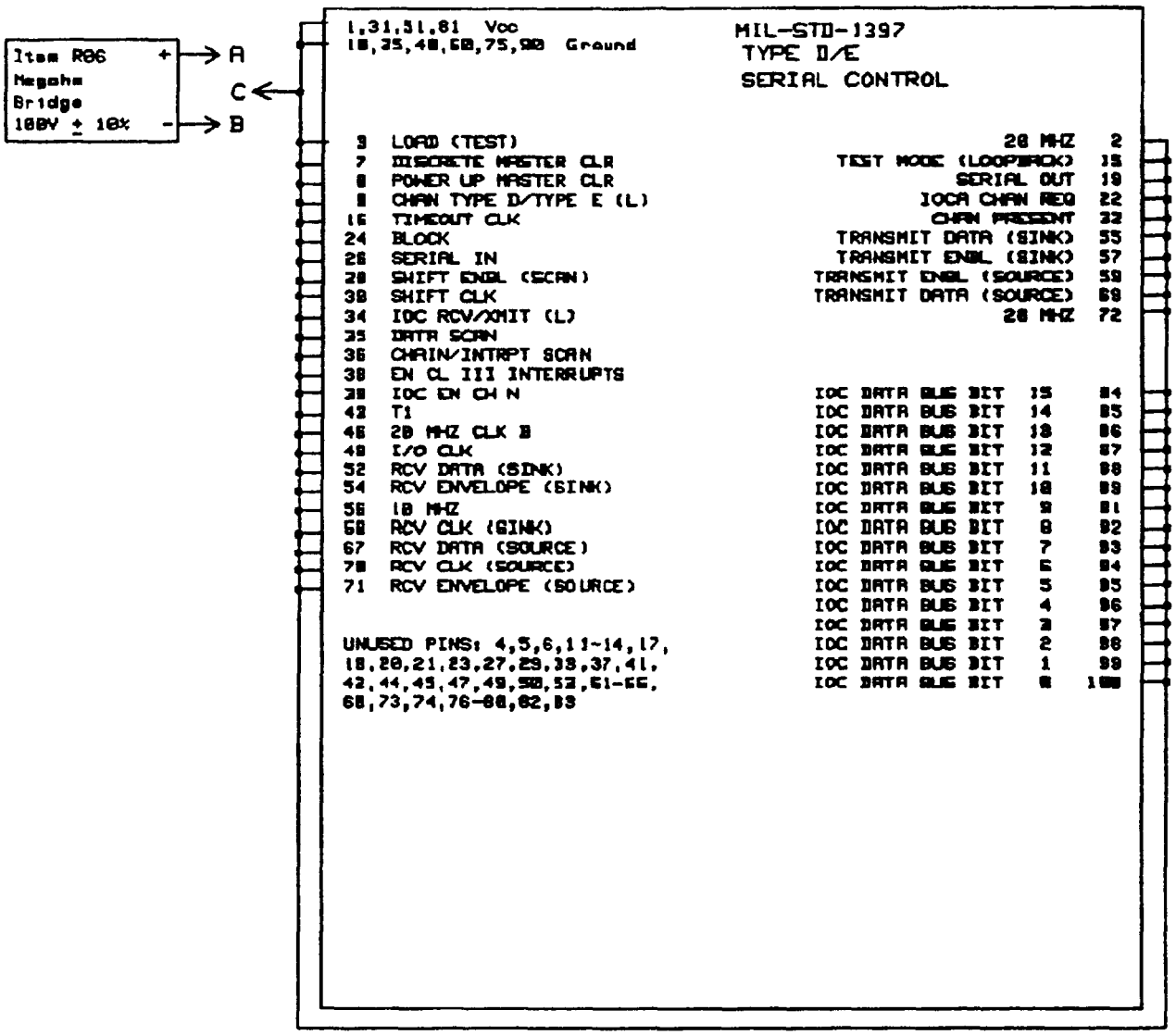


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, Vcc	1	31				
B6	Continuity, Vcc	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B6

2. The ohmmeter used shall be item R07.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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1,31,51,81 Vcc
10,25,40,60,75,90 Ground

MIL-STD-1397
TYPE D/E
SERIAL CONTROL

3	LOAD (TEST)	20 MHZ	2
7	DISCRETE MASTER CLR	TEST MODE (LOOPBACK)	15
8	POWER UP MASTER CLR	SERIAL OUT	19
9	CHAN TYPE D/TYPE E (L)	IOCA CHAN REQ	22
16	TIMEOUT CLK	CHAN PRESENT	32
24	BLOCK	TRANSMIT DATA (SINK)	55
26	SERIAL IN	TRANSMIT ENBL (SINK)	57
28	SHIFT ENBL (SCAN)	TRANSMIT ENBL (SOURCE)	59
30	SHIFT CLK	TRANSMIT DATA (SOURCE)	69
34	IOC RCV/XMIT (L)	20 MHZ	72
35	DATA SCAN		
36	CHAIN/INTRPT SCAN		
38	EN CL III INTERRUPTS		
39	IOC EN CH N	IOC DATA BUS BIT 15	84
43	T1	IOC DATA BUS BIT 14	85
46	20 MHZ CLK B	IOC DATA BUS BIT 13	86
48	I/O CLK	IOC DATA BUS BIT 12	87
52	RCV DATA (SINK)	IOC DATA BUS BIT 11	88
54	RCV ENVELOPE (SINK)	IOC DATA BUS BIT 10	89
56	10 MHZ	IOC DATA BUS BIT 9	91
58	RCV CLK (SINK)	IOC DATA BUS BIT 8	92
67	RCV DATA (SOURCE)	IOC DATA BUS BIT 7	93
70	RCV CLK (SOURCE)	IOC DATA BUS BIT 6	94
71	RCV ENVELOPE (SOURCE)	IOC DATA BUS BIT 5	95
		IOC DATA BUS BIT 4	96
		IOC DATA BUS BIT 3	97
	UNUSED PINS: 4,5,6,11-14,17,	IOC DATA BUS BIT 2	98
	18,20,21,23,27,29,33,37,41,	IOC DATA BUS BIT 1	99
	42,44,45,47,49,50,53,61-66,	IOC DATA BUS BIT 0	100
	68,73,74,76-80,82,83		

FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	2.68 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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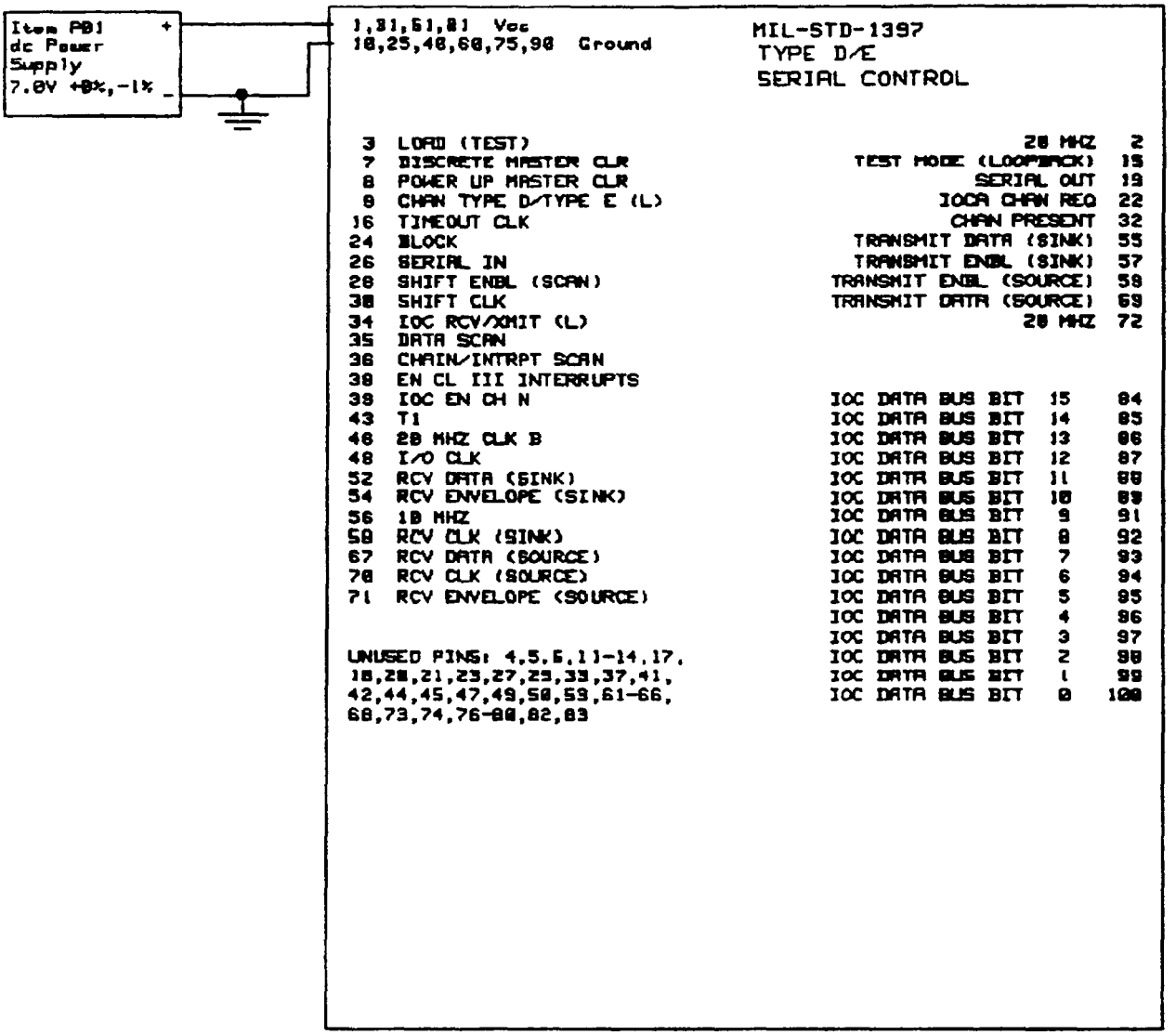


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	LOAD(TEST)	3	5.5	-0.12	0.22	mA	1-3
D2	IOC MCLR	7	5.5	0.0	0.1	mA	1-3
D3	PWR MCLR	8					
D4	CHTYPE D/E	9					
D5	TMOUT CK	16	5.5	0.0	0.1	mA	1-3
D6	BLOCK	24	5.5	-0.12	0.22	mA	1-3
D7	SERIN	26	5.5	-0.12	1.11	mA	1-3
D8	SHIFT EN	28	5.5	-0.12	0.22	mA	1-3
D9	SHIFT CK	30	5.5	-0.12	0.22	mA	1-3
D10	XMIT	34	5.5	0.0	0.1	mA	1-3
D11	DAT SCN	35					
D12	CHN/INT SCN	36					
D13	ENCLIII INT	38					
D14	IOC ENCH N	39					
D15	T1	43	5.5	0.0	0.1	mA	1-3
D16	20MHZ IN	46	5.5	0.0	1.0	mA	1-3
D17	I/O CK	48	5.5	0.0	4.0	mA	1-3
D18	RCVDAT(SNK)	52	5.5	0.0	1.0	mA	1-3
D19	RCVEN(SNK)	54					
D20	10MHZ IN	56	5.5	0.0	1.0	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	RCVCK(SNK)	58	5.5	0.0	1.0	mA	1-3
D22	RCVDAT(SRC)	67					
D23	RCVCK(SRC)	70					
D24	RCVEN(SRC)	71	5.5	0.0	1.0	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D24
Power supply current limit	2.23 A	NA	D1-D24
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D24
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 D1-D24 D1-D24
Input current measurement	NA NA NA NA NA NA NA NA NA	± 17.0 uA ± 5.0 uA ± 17.0 uA ± 61.5 uA ± 17.0 uA ± 5.0 uA ± 50.0 uA ± 200.0 uA ± 50.0 uA	D1 D2-D5 D6 D7 D8,D9 D10-D15 D16 D17 D18-D24

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	20MHZ OUT 1	2	0.0	-100.0	-40.0	mA	1-4
E2	TSTMDE(LPB)	15	0.0	-220.0	-60.0	mA	1-4
E3	SEROUT	19	0.0	-100.0	-40.0	mA	1-4
E4	IOCACHRQ	22	0.0	-150.0	-60.0	mA	1-4
E5	XMITDT(SNK)	55					
E6	XMITEN(SNK)	57					
E7	XMITEN(SRC)	59					
E8	XMITDT(SRC)	69	0.0	-150.0	-60.0	mA	1-4
E9	20MHZ OUT 2	72	0.0	-100.0	-40.0	mA	1-4
E10	IOCDAT 15	84	0.0	-150.0	-60.0	mA	1-4
E11	IOCDAT 14	85					
E12	IOCDAT 13	86					
E13	IOCDAT 12	87					
E14	IOCDAT 11	88					
E15	IOCDAT 10	89					
E16	IOCDAT 9	91					
E17	IOCDAT 8	92					
E18	IOCDAT 7	93					
E19	IOCDAT 6	94					
E20	IOCDAT 5	95	0.0	-150.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	IOCDAT 4	96	0.0	-150.0	-60.0	mA	1-4
E22	IOCDAT 3	97					
E23	IOCDAT 2	98					
E24	IOCDAT 1	99					
E25	IOCDAT 0	100	0.0	-150.0	-60.0	mA	1-4

NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E25
Power supply current limit	2.23 A	NA	E1-E25
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E25
Pattern voltage high, (VH)	3.0 V	± 100 mV	E1-E25
low, (VL)	0.0 V	± 100 mV	E1-E25

Source: https://assist.dla.mil -- Downloaded: 20261O-20T18:36Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output current measurement	NA	± 3.0 mA	E1
	NA	± 8.0 mA	E2
	NA	± 3.0 mA	E3
	NA	± 4.5 mA	E4-E8
	NA	± 3.0 mA	E9
	NA	± 4.5 mA	E10-E25

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	LOAD(TEST)	3	2.4	-5.73	-2.85	mA	1-3
F2	IOC MCLR	7	2.4	-0.2	0.02	mA	1-3
F3	PWR MCLR	8	2.4	-0.4	0.02	mA	1-3
F4	CHTYPE D/E	9	2.4	-0.2	0.02	mA	1-3
F5	TMOUT CK	16	2.4	-0.4	0.02	mA	1-3
F6	BLOCK	24	2.4	-5.63	-2.85	mA	1-3
F7	SERIN	26	2.4	-7.43	-2.82	mA	1-3
F8	SHIFT EN	28	2.4	-5.63	-2.85	mA	1-3
F9	SHIFT CK	30	2.4	-5.83	-2.85	mA	1-3
F10	XMIT	34	2.4	-0.6	0.02	mA	1-3
F11	DAT SCN	35					
F12	CHN/INT SCN	36	2.4	-0.6	0.02	mA	1-3
F13	ENCLIII INT	38	2.4	-0.4	0.02	mA	1-3
F14	IOC ENCH N	39	2.4	-0.6	0.02	mA	1-3
F15	T1	43	2.4	-0.6	0.02	mA	1-3
F16	20MHZ IN	46	2.4	-2.0	0.05	mA	1-3
F17	I/O CK	48	2.4	-8.0	0.2	mA	1-3
F18	RCVDAT(SNK)	52	2.4	-2.0	0.05	mA	1-3
F19	RCVEN(SNK)	54					
F20	10MHZ IN	56	2.4	-2.0	0.05	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	RCVCK(SNK)	58	2.4	-2.0	0.05	mA	1-3
F22	RCVDAT(SRC)	67					
F23	RCVCK(SRC)	70					
F24	RCVEN(SRC)	71	2.4	-2.0	0.05	mA	1-3
F25	IOCDAT 15	84	2.4	-0.25	0.07	mA	1-3
F26	IOCDAT 14	85					
F27	IOCDAT 13	86					
F28	IOCDAT 12	87					
F29	IOCDAT 11	88					
F30	IOCDAT 10	89					
F31	IOCDAT 9	91					
F32	IOCDAT 8	92					
F33	IOCDAT 7	93					
F34	IOCDAT 6	94					
F35	IOCDAT 5	95					
F36	IOCDAT 4	96					
F37	IOCDAT 3	97					
F38	IOCDAT 2	98					
F39	IOCDAT 1	99					
F40	IOCDAT 0	100	2.4	-0.25	0.07	mA	1-3

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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F40
Power supply current limit	2.23 A	NA	F1-F40
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F40
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	F1-F40
low, (VL)	0.0 V	± 100 mV	F1-F40
Input current measurement	NA	± 144 uA	F1
	NA	± 11 uA	F2
	NA	± 21 uA	F3
	NA	± 11 uA	F4
	NA	± 21 uA	F5
	NA	± 139 uA	F6
	NA	± 230 uA	F7
	NA	± 139 uA	F8
	NA	± 149 uA	F9
	NA	± 31 uA	F10-F12
	NA	± 21 uA	F13
	NA	± 31 uA	F14,F15
	NA	± 102 uA	F16
	NA	± 410 uA	F17
	NA	± 102 uA	F18-F24
	NA	± 16 uA	F25-F40

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	LOAD(TEST)	3	0.4	-5.73	-2.85	mA	1-3
G2	IOC MCLR	7	0.4	-0.2	0.02	mA	1-3
G3	PWR MCLR	8	0.4	-0.4	0.02	mA	1-3
G4	CHTYPE D/E	9	0.4	-0.2	0.02	mA	1-3
G5	TMOUT CK	16	0.4	-0.4	0.02	mA	1-3
G6	BLOCK	24	0.4	-5.63	-2.85	mA	1-3
G7	SERIN	26	0.4	-7.43	-2.82	mA	1-3
G8	SHIFT EN	28	0.4	-5.63	-2.85	mA	1-3
G9	SHIFT CK	30	0.4	-5.83	-2.85	mA	1-3
G10	XMIT	34	0.4	-0.6	0.02	mA	1-3
G11	DAT SCN	35					
G12	CHN/INT SCN	36	0.4	-0.6	0.02	mA	1-3
G13	ENCLIII INT	38	0.4	-0.4	0.02	mA	1-3
G14	IOC ENCH N	39	0.4	-0.6	0.02	mA	1-3
G15	T1	43	0.4	-0.6	0.02	mA	1-3
G16	20MHZ IN	46	0.4	-2.0	0.05	mA	1-3
G17	I/O CK	48	0.4	-8.0	0.2	mA	1-3
G18	RCVDAT(SNK)	52	0.4	-2.0	0.05	mA	1-3
G19	RCVEN(SNK)	54					
G20	10MHZ IN	56	0.4	-2.0	0.05	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	RCVCK(SNK)	58	0.4	-2.0	0.05	mA	1-3
G22	RCVDAT(SRC)	67					
G23	RCVCK(SRC)	70					
G24	RCVEN(SRC)	71	0.4	-2.0	0.05	mA	1-3
G25	IOCDAT 15	84	0.4	-0.25	0.07	mA	1-3
G26	IOCDAT 14	85					
G27	IOCDAT 13	86					
G28	IOCDAT 12	87					
G29	IOCDAT 11	88					
G30	IOCDAT 10	89					
G31	IOCDAT 9	91					
G32	IOCDAT 8	92					
G33	IOCDAT 7	93					
G34	IOCDAT 6	94					
G35	IOCDAT 5	95					
G36	IOCDAT 4	96					
G37	IOCDAT 3	97					
G38	IOCDAT 2	98					
G39	IOCDAT 1	99					
G40	IOCDAT 0	100	0.4	-0.25	0.07	mA	1-3

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NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G40
Power supply current limit	2.23 A	NA	G1-G40
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G40
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 G1-G40 G1-G40
Input current measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	± 144 uA ± 11 uA ± 21 uA ± 11 uA ± 21 uA ± 139 uA ± 230 uA ± 139 uA ± 149 uA ± 31 uA ± 21 uA ± 31 uA ± 102 uA ± 410 uA ± 102 uA ± 16 uA	G1 G2 G3 G4 G5 G6 G7 G8 G9 G10-G12 G13 G14,G15 G16 G17 G18-G24 G25-G40

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	20MHZ OUT 1	2	-0.4	2.4	4.5	V	1-4
H2	TSTMDE(LPB)	15	-0.16	2.4	4.5	V	1-4
H3	SEROUT	19	-0.4	2.4	4.5	V	1-4
H4	IOCACHRQ	22					
H5	XMITDT(SNK)	55					
H6	XMITEN(SNK)	57					
H7	XMITEN(SRC)	59					
H8	XMITDT(SRC)	69					
H9	20MHZ OUT 2	72					
H10	IOCDAT 15	84					
H11	IOCDAT 14	85					
H12	IOCDAT 13	86					
H13	IOCDAT 12	87					
H14	IOCDAT 11	88					
H15	IOCDAT 10	89					
H16	IOCDAT 9	91					
H17	IOCDAT 8	92					
H18	IOCDAT 7	93					
H19	IOCDAT 6	94					
H20	IOCDAT 5	95	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	IOCDAT 4	96	-0.4	2.4	4.5	V	1-4
H22	IOCDAT 3	97					
H23	IOCDAT 2	98					
H24	IOCDAT 1	99					
H25	IOCDAT 0	100	-0.4	2.4	4.5	V	1-4

NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H25
Power supply current limit	2.23 A	NA	H1-H25
Forcing current, (Ioh)	-0.4 mA -0.16 mA -0.4 mA	± 40 uA ± 16 uA ± 40 uA	H1 H2 H3-H25
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H25 H1-H25
Output voltage measurement	NA	± 105 mV	H1-H25

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

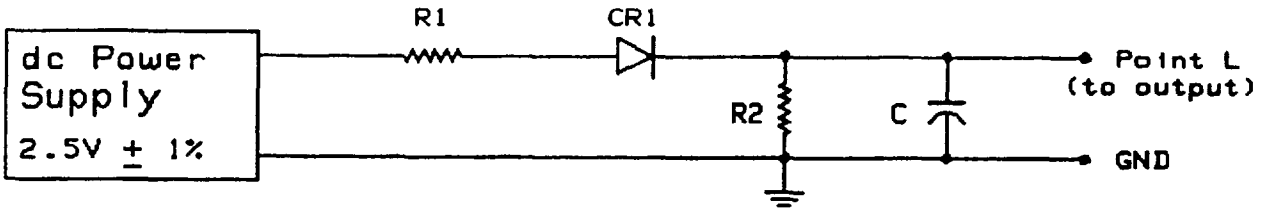
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are specified below.

4. The value of R2 for each output load shall be as specified below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
2,19,22,55,57, 59,69,72,84,85, 86,87,88,89,91, 92,93,94,95,96, 97,98,99,100	6200	6.0	0.5
15	16000	1.6	0.5
32	NA	1.6	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J1	20MHZ OUT 1	2	6.0	0.0	0.5	V	1-4
J2	TSTMDE(LPB)	15	1.6	0.0	0.5	V	1-4
J3	SEROUT	19	6.0	0.0	0.5	V	1-4
J4	IOCACHRQ	22	6.0	0.0	0.5	V	1-4
J5	CH PRES	32	1.6	0.0	0.5	V	1-4
J6	XMITDT(SNK)	55	6.0	0.0	0.5	V	1-4
J7	XMITEN(SNK)	57					
J8	XMITEN(SRC)	59					
J9	XMITDT(SRC)	69					
J10	20MHZ OUT 2	72					
J11	IOCDAT 15	84					
J12	IOCDAT 14	85					
J13	IOCDAT 13	86					
J14	IOCDAT 12	87					
J15	IOCDAT 11	88					
J16	IOCDAT 10	89					
J17	IOCDAT 9	91					
J18	IOCDAT 8	92					
J19	IOCDAT 7	93					
J20	IOCDAT 6	94	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J21	IOCDAT 5	95	6.0	0.0	0.5	V	1-4
J22	IOCDAT 4	96					
J23	IOCDAT 3	97					
J24	IOCDAT 2	98					
J25	IOCDAT 1	99					
J26	IOCDAT 0	100	6.0	0.0	0.5	V	1-4

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J26
Power supply current limit	2.23 A	NA	J1-J26
Forcing current, (Iol)	6.0 mA 1.6 mA 6.0 mA 1.6 mA 6.0 mA	± 600 uA ± 160 uA ± 600 uA ± 160 uA ± 600 uA	J1 J2 J3,J4 J5 J6-J26

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:36Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 J1-J26 J1-J26
Output voltage measurement	NA	± 25 mV	J1-J26

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
K	CH PRES	32	5.5	0.0	0.25	mA	1-3

NOTES:

1. Output leakage current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K
Power supply current limit	2.23 A	NA	K
Forcing voltage, (Voh)	5.5 V	± 50 mV	K
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	K K
Output current measurement	NA	± 12.5 uA	K

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	IOC MCLR	7	-18.0	-1.5	0.0	V	1-3
L2	PWR MCLR	8					
L3	CHTYPE D/E	9					
L4	TMOUT CK	16	-18.0	-1.5	0.0	V	1-3
L5	XMIT	34	-18.0	-1.2	0.0	V	1-3
L6	DAT SCN	35					
L7	CHN/INT SCN	36	-18.0	-1.2	0.0	V	1-3
L8	ENCLIII INT	38	-18.0	-1.5	0.0	V	1-3
L9	IOC ENCH N	39	-18.0	-1.2	0.0	V	1-3
L10	T1	43					
L11	20MHZ IN	46					
L12	I/O CK	48					
L13	RCVDAT(SNK)	52					
L14	RCVEN(SNK)	54					
L15	10MHZ IN	56					
L16	RCVCK(SNK)	58					
L17	RCVDAT(SRC)	67					
L18	RCVCK(SRC)	70					
L19	RCVEN(SRC)	71	-18.0	-1.2	0.0	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L19
Power supply current limit	NA	NA	L1-L19
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L19
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L19 L1-L19
Input voltage measurement	NA NA NA NA	± 75 mV ± 60 mV ± 75 mV ± 60 mV	L1-L4 L5-L7 L8 L9-L19

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	4.3	8.0	uF	1-4
M2	PSFC	51	3.7	6.9	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA NA	± 185 nF ± 160 nF	M1 M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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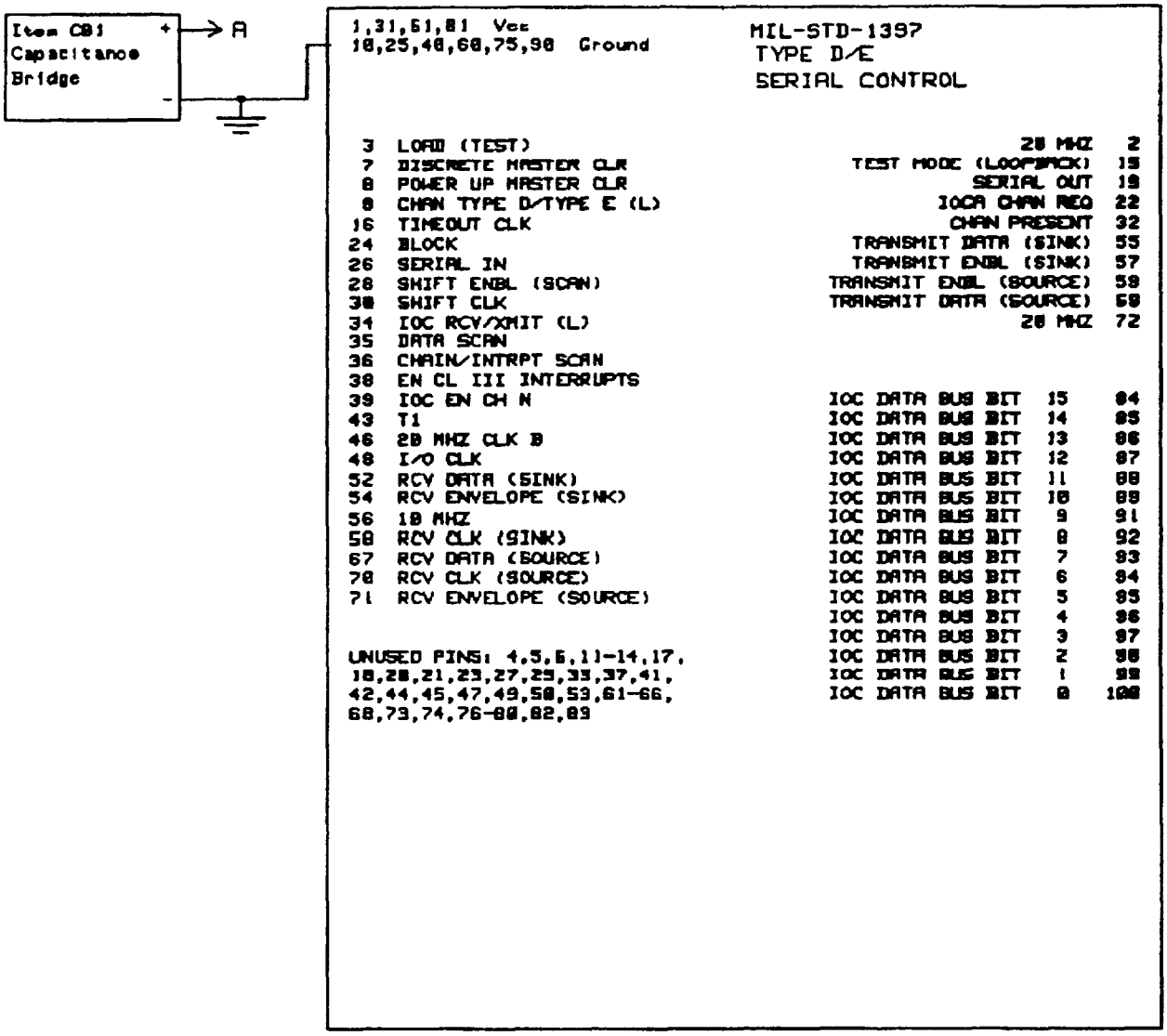


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31, 51,81	342	1487	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	2.23 A	NA	N
Pattern voltage high, (VH)	3.0 V	± 100 mV	N
low, (VL)	0.0 V	± 100 mV	N
Power supply current measurement	NA	± 57.2 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (225 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (225 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	2.23 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 20 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

3. Abbreviations used are defined in 6.1.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (225 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (225 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (225 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (225 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	2.23 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

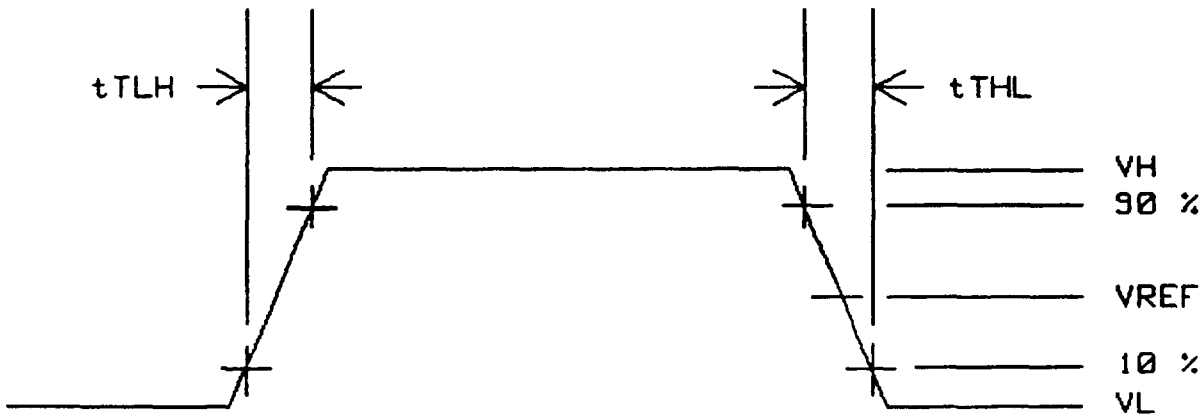
(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. I/O waveforms are as referenced on figure 11. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	10MHZ IN to XMITEN(SRC)	-	-	-	90	90	90	nS	11
1	tpd	10MHZ IN to XMITDT(SRC)	-	-	-	90	90	90	nS	11
1	tpd	10MHZ IN to XMITEN(SNK)	-	-	-	90	90	90	nS	11
1	tpd	10MHZ IN to XMITDT(SNK)	-	-	-	90	90	90	nS	11
2	tpd	10MHZ IN to SEROUT	-	-	-	85	85	85	nS	11
3	tpd	CHN/INT SCN to IOCACHRQ	-	-	-	69	69	69	nS	11
3	tpd	DAT SCN to IOCACHRQ	-	-	-	69	69	69	nS	11

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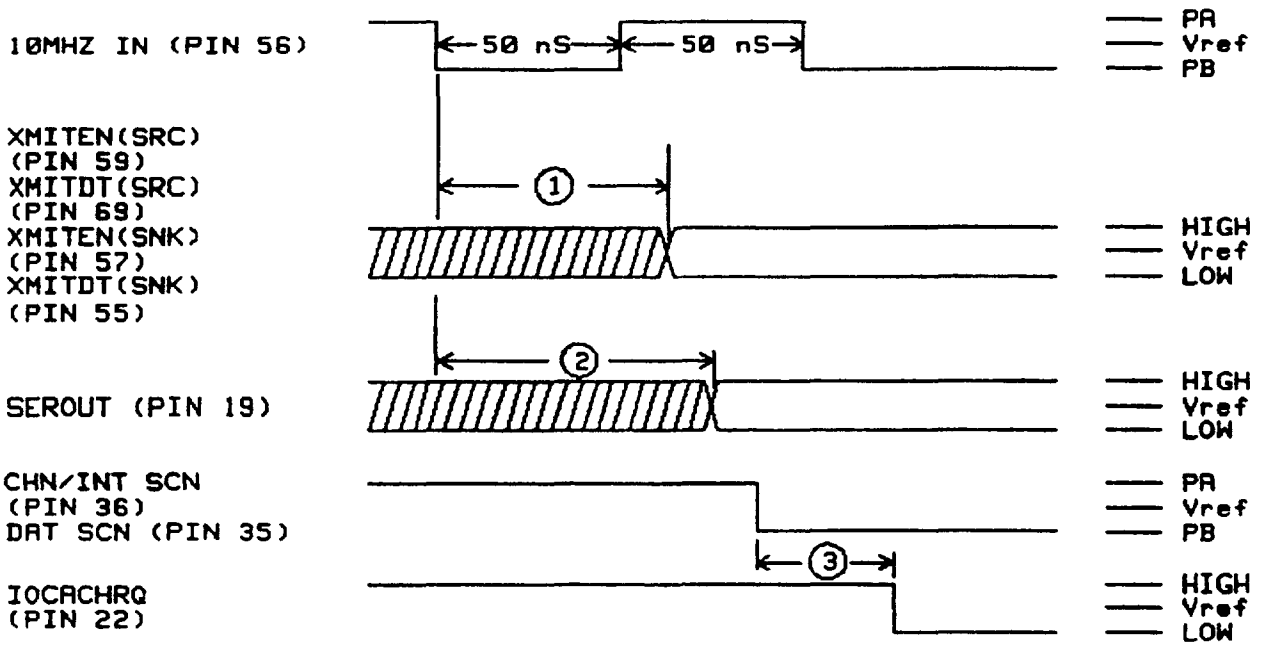


FIGURE 11. Dynamic waveforms for table XVII.

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TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	LOAD(TEST)	3	4	75	pF	1-4
R2	IOC MCLR	7	2	75	pF	1-4
R3	PWR MCLR	8				
R4	CHTYPE D/E	9				
R5	TMOUT CK	16	2	75	pF	1-4
R6	BLOCK	24	4	75	pF	1-4
R7	SERIN	26	4	100	pF	1-4
R8	SHIFT EN	28	4	100	pF	1-4
R9	SHIFT CK	30	4	75	pF	1-4
R10	XMIT	34	2	75	pF	1-4
R11	DAT SCN	35				
R12	CHN/INT SCN	36				
R13	ENCLIII INT	38				
R14	IOC ENCH N	39				
R15	T1	43	2	75	pF	1-4
R16	20MHZ IN	46	2	100	pF	1-4
R17	I/O CK	48	8	150	pF	1-4
R18	RCVDAT(SNK)	52	2	75	pF	1-4
R19	RCVEN(SNK)	54				
R20	10MHZ IN	56	2	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	RCVCK(SNK)	58	2	75	pF	1-4
R22	RCVDAT(SRC)	67				
R23	RCVCK(SRC)	70				
R24	RCVEN(SRC)	71	2	75	pF	1-4
Bidirectionals						
R25	IOCDAT 15	84	4	75	pF	1-4
R26	IOCDAT 14	85				
R27	IOCDAT 13	86				
R28	IOCDAT 12	87				
R29	IOCDAT 11	88				
R30	IOCDAT 10	89				
R31	IOCDAT 9	91				
R32	IOCDAT 8	92	4	75	pF	1-4
R33	IOCDAT 7	93	4	100	pF	1-4
R34	IOCDAT 6	94	4	125	pF	1-4
R35	IOCDAT 5	95	4	100	pF	1-4
R36	IOCDAT 4	96				
R37	IOCDAT 3	97				
R38	IOCDAT 2	98				
R39	IOCDAT 1	99				
R40	IOCDAT 0	100	4	100	pF	1-4

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NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R40
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R40
Capacitance measurement	NA	± 3.55 pF	R1
	NA	± 3.65 pF	R2-R5
	NA	± 3.55 pF	R6
	NA	± 4.8 pF	R7,R8
	NA	± 3.55 pF	R9
	NA	± 3.65 pF	R10-R15
	NA	± 4.9 pF	R16
	NA	± 7.1 pF	R17
	NA	± 3.65 pF	R18-R24
	NA	± 3.55 pF	R25-R32
	NA	± 4.8 pF	R33
	NA	± 6.05 pF	R34
	NA	± 4.8 pF	R35-R40

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 12.

4. Abbreviations used are defined in 6.1.

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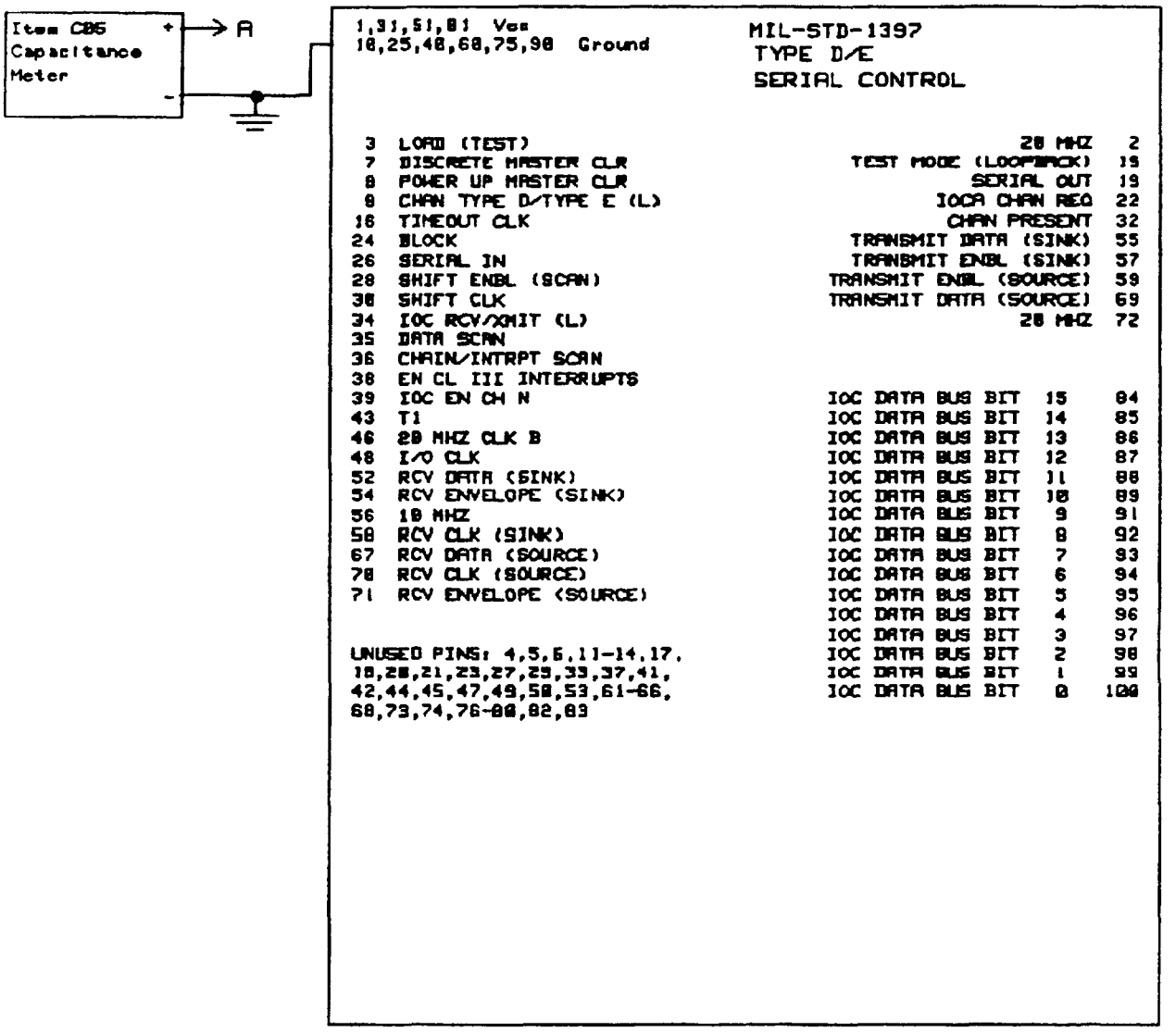


FIGURE 12. Input and output capacitance test circuit for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T1	PSTA	1	0	400	mV	1-4
T2	PSTA	51	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	2.23 A	NA	T1,T2
Pattern voltage high, (VH)	3.0 V	± 100 mV	T1,T2
low, (VL)	0.0 V	± 100 mV	T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 13 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 14.

4. Abbreviations used are defined in 6.1.

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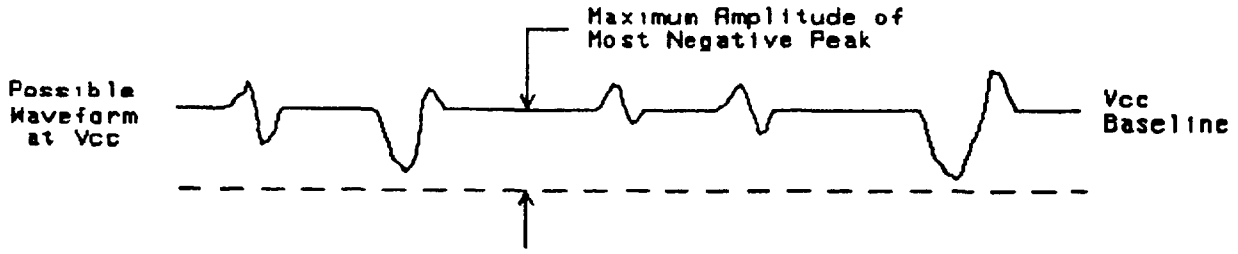
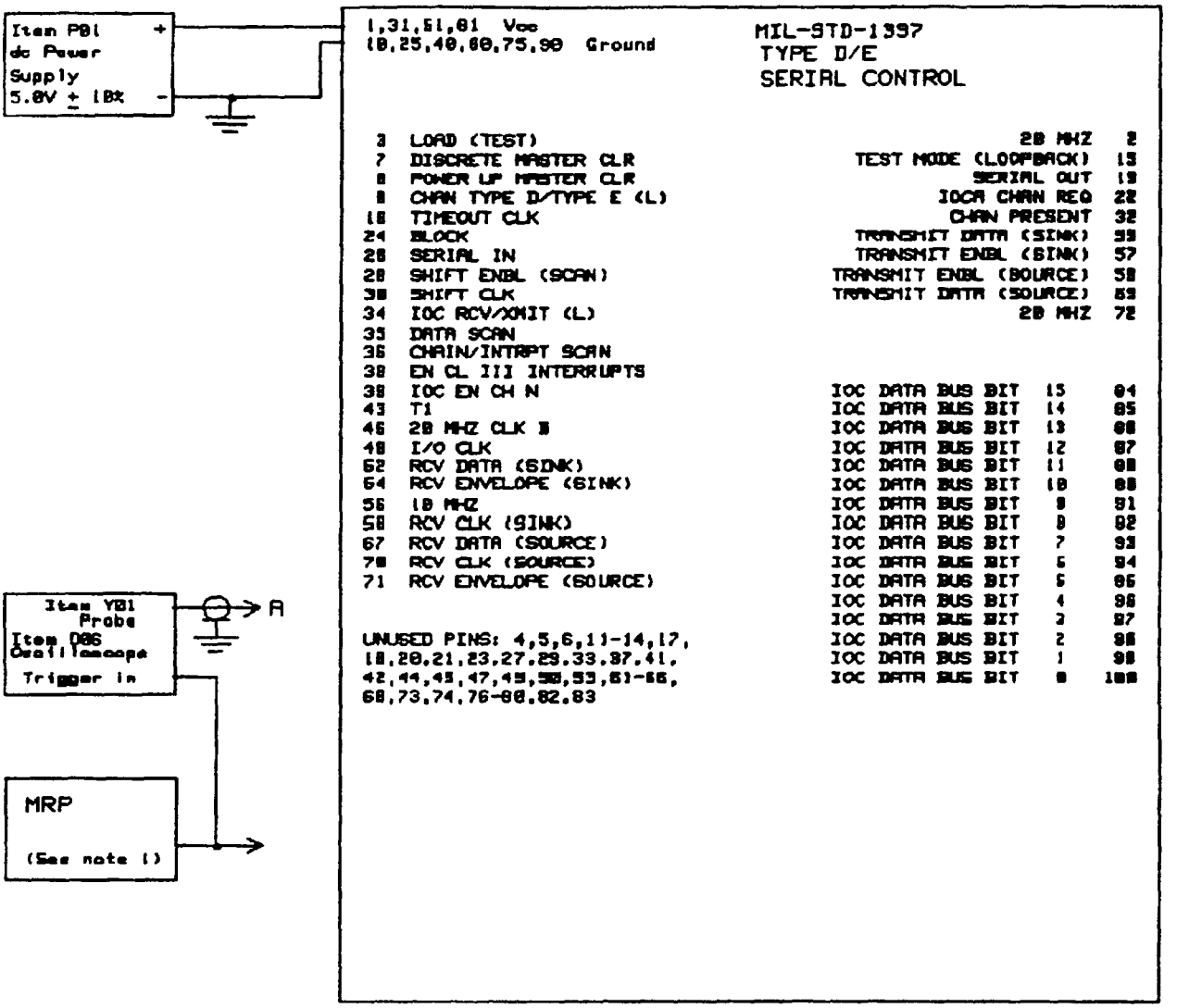


FIGURE 13. Power supply transient amplitude waveform for table XIX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 14. Power supply transient amplitude test circuit for table XIX.

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	MIL-M-28787/528	

3. DOCUMENT TITLE: Modules, Standard Electronic, MIL-STD-1397 Type D/E Serial Control Module, Key Code AJK

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
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(2) AUTOVON: 88-482-1528

c. ADDRESS (Include Zip Code): Commanding Officer
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